

Bullet Farm

A transaction processor around powerful coding agents

Executive brief — Stage-1 architecture and component assurance — 25 August 2026

Decision in one sentence. Treat an agent-produced software change like a recoverable transaction: bind who may write, the exact bytes being verified, and the exact remote effect before allowing integration.

Why this is powerful

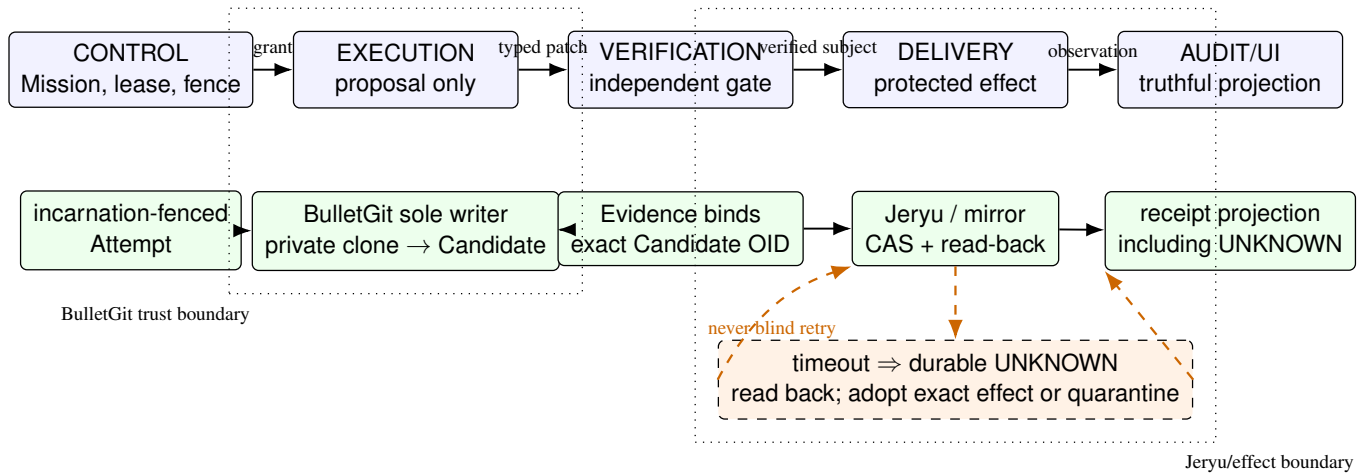
Today an agent can reason brilliantly and still lose work for mundane reasons: two sessions share a checkout, a retry duplicates a workflow, CI approves a branch that moved, or a dashboard guesses that silence means success. A sandbox, worktree, reviewer, and merge queue each help, but none alone answers all four questions: *who owns the write, which incarnation is current, which exact bytes were proved, and did the remote mutation really happen?*

Bullet Farm puts those answers outside the model. Providers may explore and propose. They cannot grant themselves repository, verification, or merge authority. The repository remains sovereign even when models and vendors are swapped.

1. **Concurrency becomes explicit.** A lease plus an incarnation fence means a process that wakes after reclaim is stale, even if it still runs.
2. **Git identity becomes exact.** One BulletGit writer applies typed patches in private Attempt clones and mints an immutable Candidate from exact base/head/tree/provenance subjects.
3. **Evidence cannot float.** An independent verifier proves that Candidate, not a moving branch, a transcript, or a model's "done" statement.
4. **Timeout is survivable.** Durable effect intent plus read-back distinguishes "applied," "not applied," and honest UNKNOWN; the system never blindly retries an ambiguous push.
5. **The UI cannot manufacture truth.** Five-plane projections retain stale, contradictory, failed, and unknown outcomes instead of painting them green.

Evidence boundary. Snapshot 20260825-stage1 binds Hub f33750c14112, Kernel 419d8a6832ee, BulletGit 236f4efe2b82, and Portal 95108e37d643. It reports 26 blocked of 26 evaluated release gates and no connected transaction proof.

The binding chain



The design composes eight bindings:

incarnation fence → sole writer → private Attempt clone → exact-OID Candidate → independent Evidence → durable effect
intent → read-back/adoption → truthful projection

A failure, recovered without guessing

Suppose a protected-ref update times out after the forge may have accepted it. A conventional retry loop often assumes failure and sends the mutation again. Bullet Farm has already committed one logical effect identity, expected-old OID, intended-new OID, and Evidence root. It records UNKNOWN and reads the protected ref back:

Observed state	Required action
Ref equals intended OID	Adopt the already-applied effect and seal the read-back receipt.
Ref equals expected- old OID	Retry the same logical effect identity under the current fence.
Ref equals neither	Quarantine divergence; preserve bytes and require adjudication.
No trustworthy obser- vation	Remain UNKNOWN; do not infer success or failure.

This is why Bullet Farm is a transaction processor rather than another agent loop. The important unit is not a chat session. It is the recoverable journey from authority to exact Candidate, Evidence, protected effect, and observation.

What adjacent systems contribute

Gas Town and Gas City provide strong orchestration, durable work, multiple runtimes, and merge-queue operations. DeepSeek Harness supplies a composable plugin kernel plus an experimental Teams seam. Omnigent provides cross-vendor sessions, Polly's parallel worktrees and review, OS sandboxes, and an L7 credential proxy. Bullet Farm should retain those product lessons. Its narrower claim is that the reviewed pinned public contracts do not document this same end-to-end authority/evidence/effect chain.

Linked Git worktrees have per-worktree HEAD and index while sharing the object database, common metadata, and most refs. They are useful workspace isolation, not separate repository principals.

Implemented now / not yet proved

Prominent limit. Every positive Bullet result below is component evidence. TRANSACTION_PROOF, live-provider/forged proof, and release proof are absent. The release inventory is BLOCKED: 26 blocked of 26 evaluated gates.

Boundary	Implemented now	Not yet proved
Control authority	DB-clock lease/fence, atomic command/outbox, authenticated loopback ingress	Signed connected authority from Kernel through Runner and BulletGit
Provider boundary	Offline protocol subsets, admission and egress components, policy-first refusal	Any live provider receipt; live policy remains disabled in the evidence subject
Repository writer	Fail-closed BulletGit gateway, private-clone/preservation components, strict manifests	Positive online sole-writer transaction
Candidate/Evidence	Exact-subject fixture gate and bounded verifier transport	Production reconstruction, executable identity for all gates, multi-gate aggregation
Effects	Ambiguity model and local bare-forged components	Authenticated expected-old-OID push, read-back, adoption, and protected integration
Audit/Portal	Generated DTOs, exact command reconciliation, watermark projections, explicit UNKNOWN	Packaged complete product and every designed ledger subject
Formal assurance	2 bounded safety models, PASS at pinned state counts	Liveness, OS/Git/provider/forged composition, or a product proof
Release	Negative inventory; audit 58/raw 58, 10 caps, 29 hard findings	Quality floor, portable audit, packages, SBOM/provenance, signing, install/recovery, live receipts

How to read maturity

Designed	Reviewed rule; executable path may not exist.
Component-proved	Exact committed component plus mapped receipt; no cross-plane inference.
Transaction-proved	One exact connected offline five-plane transaction with independent receipts.
Live/release-proved	Admitted live provider/forged transaction, or signed package/install/recovery proof.

The snapshot reports doctor BLOCKED, fast PASS, demo PASS, and transaction proof ABSENT. Demo success is not silently relabeled as a transaction. A timeout, skip, unsupported path, missing receipt, or dirty-tree result cannot be promoted through prose.

The practical consequence

Bullet Farm is powerful as a control architecture now, but not yet proved as a connected product. That distinction is a feature: it makes investment and risk decisions against a visible boundary instead of a polished demonstration.

Publication, forge topology, and next decisions

Two publication stages

Stage 1 — now. Publish the architecture and component-assurance preprint, the four-page brief, bibliography, vector figure, JSON evidence snapshot, generated macros, and deterministic preflight. Make no comparative performance claim.

Stage 2 — later. Only after a signed connected TRANSACTION_PROOF, run a matched receipt-bearing corpus against pinned systems. Record exact tasks, subjects, models, budgets, retries, costs, collisions, duplicate effects, unknowns, recovery, and loss. Only this stage may support “faster,” “cheaper,” “safer,” or “more successful.”

Frozen V1 topology and blocked proposals

- V1 source distribution is Jeryu-only from immutable signed tags. The manifest admits no authenticated Jeryu URL and assumes no public mirror.
- The only permitted local Jeryu family is /home/ubuntu/jain-split/jeryu-split; its branded hub maps independent component repositories. Never recreate /home/ubuntu/jeryu-split.
- `git.neverhuman.org`, `github.com/neverhuman/jeryu`, and `github.com/neverhuman/jankurai` are unratified, operator-blocked future proposals—not current authority, authenticated endpoints, or public front doors.
- Any later public mirror remains a separately configured effect adapter. It requires expected-old-OID CAS, read-back, divergence quarantine, and separately proved backups.
- Jankurai stays the existing monorepo at /home/ubuntu/jankurai. No public front door is assumed; preservation must precede any mirror decision.

The local-family rule is frozen; the endpoint and mirror proposals are not. Operator ratification plus TLS, auth, protection, deployment, mirror, backup, and restoration receipts are required before any proposed name becomes authority or a public front door.

Immediate decisions

1. Fund the connected five-plane transaction proof before comparative marketing.
2. Publish immutable signed bullet-wire only through an operator-ratified, authenticated Jeryu endpoint.
3. Select and prove protected Jeryu integration plus the ratified endpoint’s operational controls.
4. Finish deterministic packages, SBOM/provenance, schema-3 installation, and portable Jankurai audit evidence.
5. Resolve Jeryu release-authority document drift and preserve Jankurai history before any mirror decision.
6. Publish immutable paper/source permalinks only after strict preflight is green from clean subjects.

Bottom line. More capable models increase the value of external authority, exact evidence, and recoverable effects. Bullet Farm has a defensible transaction architecture and meaningful components. It does not yet have the connected proof needed to claim a finished product or measured superiority.